Step 1 : ENTER YOLO FOLDER (FROM GOOGLE DRIVE)

```
!pip install uv
!uv pip install --system ultralytics
```

```
Collecting uv
  Downloading uv-0.11.6-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metada
Downloading uv-0.11.6-py3-none-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (25.0 MB
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 25.0/25.0 MB 65.6 MB/s eta 0:00:00
Installing collected packages: uv
Successfully installed uv-0.11.6
Using Python 3.12.13 environment at: /usr
Resolved 53 packages in 1.01s
Prepared 2 packages in 260ms
Installed 2 packages in 29ms
 + ultralytics==8.4.37
 + ultralytics-thop==2.0.18
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mo
```

```
BASE_PATH = "/content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO"
```

```
!ls /content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO
```

```
data_fixed.yaml  data.yaml  test  train  valid
```

```
!ls /content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/train/images | head
```

```
000dc27f-NAJIB_MARDANLOO_MASUME_2020-07-12185357_jpg.rf.76cabac5ba504297e98502d2731728
000dc27f-NAJIB_MARDANLOO_MASUME_2020-07-12185357_jpg.rf.8ffbe5dc98f1e8997ea22f5ef5abc3
003e5f80-MOHAMADI_ATEFEH_2020-06-01173808_jpg.rf.f236525d49813bc09dfbe4b139c412ac.jpg
0043783e-Asiadar_Leila_2022-06-12141026_jpg.rf.f6ffdb23ca9d38c757b7001aa84cf1e7.jpg
00cf39c1-Karaptiyan_Robert_50yo_13032021_185908_jpg.rf.98b2e72cb9a26e75d40df97e04473ac
00cf39c1-Karaptiyan_Robert_50yo_13032021_185908_jpg.rf.b8c24267668299d5d84e878357067cc
00cf39c1-Karaptiyan_Robert_50yo_13032021_185908_jpg.rf.d93e04580a7164c9e5c7ebcd57177f6
00cf39c1-Karaptiyan_Robert_50yo_13032021_185908_jpg.rf.f0189b6853ba7ccdec9f1cd91b58dac
01b0dd74-Gazavandi_Neda_2022-05-14190158_jpg.rf.171767b3f4b8c3e99f0810c1c3d31d0b.jpg
01b0dd74-Gazavandi_Neda_2022-05-14190158_jpg.rf.93976cb5376c518410a89cbbc608a0a8.jpg
```

```
!ls /content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/train/labels | head
```

```
000dc27f-NAJIB_MARDANLOO_MASUME_2020-07-12185357_jpg.rf.76cabac5ba504297e98502d2731728
000dc27f-NAJIB_MARDANLOO_MASUME_2020-07-12185357_jpg.rf.8ffbe5dc98f1e8997ea22f5ef5abc3
003e5f80-MOHAMADI_ATEFEH_2020-06-01173808_jpg.rf.f236525d49813bc09dfbe4b139c412ac.txt
0043783e-Asiadar_Leila_2022-06-12141026_jpg.rf.f6ffdb23ca9d38c757b7001aa84cf1e7.txt
00cf39c1-Karaptiyan_Robert_50yo_13032021_185908_jpg.rf.98b2e72cb9a26e75d40df97e04473ac
00cf39c1-Karaptiyan_Robert_50yo_13032021_185908_jpg.rf.b8c24267668299d5d84e878357067cc
00cf39c1-Karaptiyan_Robert_50yo_13032021_185908_jpg.rf.d93e04580a7164c9e5c7ebcd57177f6
00cf39c1-Karaptiyan_Robert_50yo_13032021_185908_jpg.rf.f0189b6853ba7ccdec9f1cd91b58dac
```

```
01b0dd74-Gazavandi_Neda_2022-05-14190158_jpg.rf.171767b3f4b8c3e99f0810c1c3d31d0b.txt
01b0dd74-Gazavandi_Neda_2022-05-14190158_jpg.rf.93976cb5376c518410a89cbbc608a0a8.txt
```

```
from PIL import Image
import matplotlib.pyplot as plt

img_path = "/content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/train/images/000

img = Image.open(img_path)
plt.imshow(img)
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(1614.5), np.float64(839.5), np.float64(-0.5))
```

```
label_path = "/content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/train/labels/0

with open(label_path,"r") as f:
  print(f.read())
```

```
2 0.32449374984520124 0.49987499999999996 0.3218531250773993 0.4874 0.314309375232198
2 0.360325 0.5185890619047618 0.3539125003095975 0.523440625 0.3497640625386997 0.537
9 0.3569296873065016 0.52066875 0.36145625015479876 0.5220546869047619 0.361079687306
7 0.3574593752321981 0.4261765630952381 0.36384687492260065 0.4324359380952381 0.3642
7 0.3336140625386997 0.4563 0.33404062476780183 0.46608124999999995 0.336381250154798
0 0.40280781238390095 0.581884375 0.40536250030959753 0.580320311904762 0.40664062476
9 0.38739999999999997 0.5373625 0.3858546873065015 0.496178125 0.38353593746130027 0.
9 0.39177968730650153 0.4531 0.39075 0.4985453130952381 0.39075 0.5283687499999999 0.
2 0.4152234377708979 0.59985 0.42527031269349846 0.5993765630952381 0.428876562229102
2 0.4582437498452013 0.55393125 0.4546375003095975 0.554404688095238 0.45154687492260
9 0.4435609374613003 0.548725 0.4453640625386997 0.5406765630952381 0.448196874922600
2 0.48297500000000004 0.563873438095238 0.47730781238390096 0.5652937499999999 0.4726
2 0.5172374996904024 0.561032811904762 0.51595 0.567659375 0.5185265622291022 0.58280
9 0.5138890625386997 0.525054688095238 0.5110546873065016 0.496178125 0.50564531269349
9 0.5471218749226006 0.516534375 0.5437718749226006 0.4862375 0.5373312501547988 0.44
9 0.5687609374613003 0.4526265619047619 0.5685031250773994 0.5080125 0.56876093746130
9 0.5790656247678019 0.452153125 0.5754593752321981 0.5032796869047619 0.576231250154
2 0.5440296873065016 0.57996875 0.5448031250773994 0.569079688095238 0.54609062476780
2 0.5867937498452013 0.5856484369047619 0.5911734377708978 0.578075 0.591173437770897
2 0.7114796873065015 0.512746875 0.7086453126934984 0.5132203130952381 0.702720312693
1 0.7251328123839009 0.53215625 0.7150859374613003 0.5335765619047619 0.704009375232
0 0.6898406247678018 0.6154718749999999 0.6885515622291022 0.623520311904762 0.690096
1 0.6870062501547988 0.62020625 0.6852031250773994 0.6078984369047619 0.678504687306
1 0.6293 0.6292 0.623117187616099 0.617365625 0.6164203126934985 0.6112109380952381 0
1 0.5893703126934984 0.6481359380952381 0.5852484377708979 0.6358281250000001 0.57880
0 0.5553656247678018 0.6547625 0.5553656247678018 0.6486093749999999 0.55587968730650
```

```
7 0.5646390625386997 0.7873125 0.5620625003095976 0.786839063095238 0.561548437770897
2 0.41728437523219813 0.629673436904762 0.4131624996904025 0.6325140630952382 0.40955
1 0.4072359374613003 0.6665984380952381 0.4072359374613003 0.659023436904762 0.404918
1 0.37838437523219814 0.6481359380952381 0.37915625015479876 0.6396156249999999 0.3758
1 0.34540937523219817 0.6632843749999999 0.3492734377708978 0.6486093749999999 0.3503
1 0.29697812507739935 0.65145 0.3029031250773994 0.6434015619047619 0.309085937461300
7 0.39930555541795665 0.8090277773809524 0.39930555541795665 0.7881944440476191 0.385
2 0.28177812507739936 0.60553125 0.28255156222910216 0.5951171880952381 0.28049062476
9 0.3894230767801858 0.7920673071428571 0.3930288464396285 0.8016826928571429 0.39302
13 0.2851562501547988 0.5143229166666666 0.33723958328173376 0.5299479166666666 0.364
6 0.6051136365325077 0.5681818178571428 0.6178977275541795 0.546875 0.610795454489164
6 0.6235795455108359 0.5582386369047619 0.6534090910216718 0.5482954547619048 0.66477
13 0.28835227244582046 0.6789772726190476 0.3139204544891641 0.708806817857143 0.3508
```

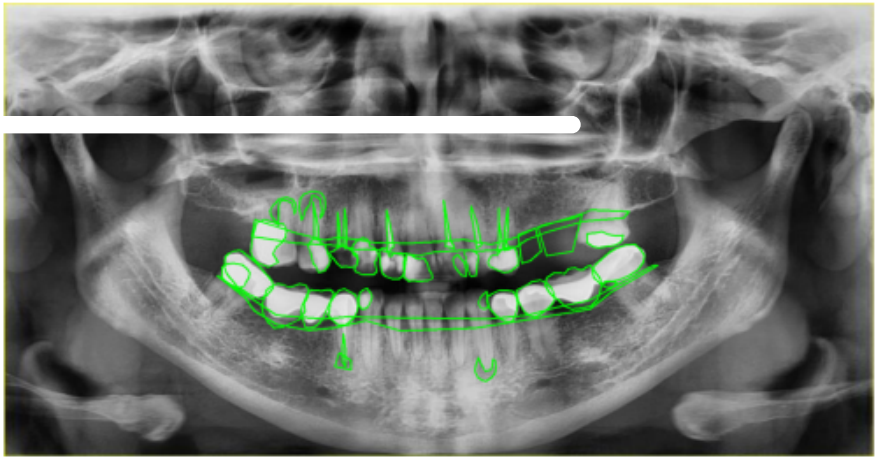
```
!cat /content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/data.yaml
```

```
# path: ../datasets/roboflow

train: /home/loki/DentalObjectDetection/data/YOLO/train/images
val: /home/loki/DentalObjectDetection/data/YOLO/valid/images
test: /home/loki/DentalObjectDetection/data/YOLO/valid/images

names:
  0: Caries
  1: Crown
  2: Filling
  3: Implant
  4: Malaligned
  5: Mandibular Canal
  6: Missing teeth
  7: Periapical lesion
  8: Retained root
  9: Root Canal Treatment
  10: Root Piece
  11: impacted tooth
  12: maxillary sinus
  13: Bone Loss
  14: Fracture teeth
  15: Permanent Teeth
  16: Supra Eruption
  17: TAD
  18: abutment
  19: attrition
  20: bone defect
  21: gingival former
  22: metal band
  23: orthodontic brackets
  24: permanent retainer
  25: post - core
  26: plating
  27: wire
  28: Cyst
  29: Root resorption
  30: Primary teeth
```

```
import cv2
import matplotlib.pyplot as plt
import numpy as np

img_path = "/content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/train/images/000
```

```
label_path = "/content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/train/labels/0(

img = cv2.imread(img_path)

h, w, _ = img.shape

with open(label_path, "r") as f:
  lines = f.readlines()

for line in lines:
  values = list(map(float, line.split()))

  cls = int(values[0]) # first value = class
  points = values[1:] #remaining = polygon points

  #Convert normalized --> pixel coords
  pts = []
  for i in range(0,len(points),2):
    x = int(points[i] * w)
    y = int(points[i+1] * h)
    pts.append([x,y])

  pts = np.array(pts,np.int32)
  pts = pts.reshape((-1,1,2))

  #Draw Polygon
  cv2.polylines(img, [pts], isClosed=True, color=(0,255,0), thickness=2)

plt.imshow(cv2.cvtColor(img,cv2.COLOR_BGR2RGB))
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(1614.5), np.float64(839.5), np.float64(-0.5))
```



```
from ultralytics import YOLO
```

```
Creating new Ultralytics Settings v0.0.6 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settir
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/(
```

```
model = YOLO("/content/drive/MyDrive/Dental_Colab_Notebooks/best.pt")
```

```
!ls /content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/test/images | head
```

```
000dc27f-NAJIB_MARDANLOO_MASUME_2020-07-12185357_jpg.rf.adab6a526a5ec0225d2ce97c33095e
000dc27f-NAJIB_MARDANLOO_MASUME_2020-07-12185357_jpg.rf.f00e39f32edbca5c4cf42f39d9b03(
003e5f80-MOHAMADI_ATEFEH_2020-06-01173808_jpg.rf.d58185833fd258d087f0e42cbfdd7c37.jpg
0043783e-Asiadar_Leila_2022-06-12141026_jpg.rf.6c123e312ec9ed9f03e1df7ecbd2c38b.jpg
00cf39c1-Karaptiyan_Robert_50yo_13032021_185908_jpg.rf.431ba12a0f81dea1ccf064d488dc5f5
045b3b84-PIRI_NAJME_2020-02-18184406_jpg.rf.b561dab76cb538a94536944cff81ef1c.jpg
05e07869-Rahimi_Moslem_2022-06-12141630_jpg.rf.4b818460c79b1cebdf4a41deed2b72d5.jpg
066c800c-Norozi_syavazan_Atefeh_2022-06-12142626_jpg.rf.87d0e87af0a6dbafa70ac290e820c3
06aceb48-Rahimi_Arefeh_2022-06-12142023_jpg.rf.2928cac7315c53b134a6872e9f5f35ac.jpg
07246a96-DIRIKVAND_REZA_2020-06-29174111_jpg.rf.d26b7b3a95131fa2ac8f4509f1c4a3bf.jpg
```

```
img_path = "/content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/test/images/000d(
```

```
results = model.predict(
    source = img_path,
    save = True, #save output image
    conf=0.25 # confidence threshold
)
```

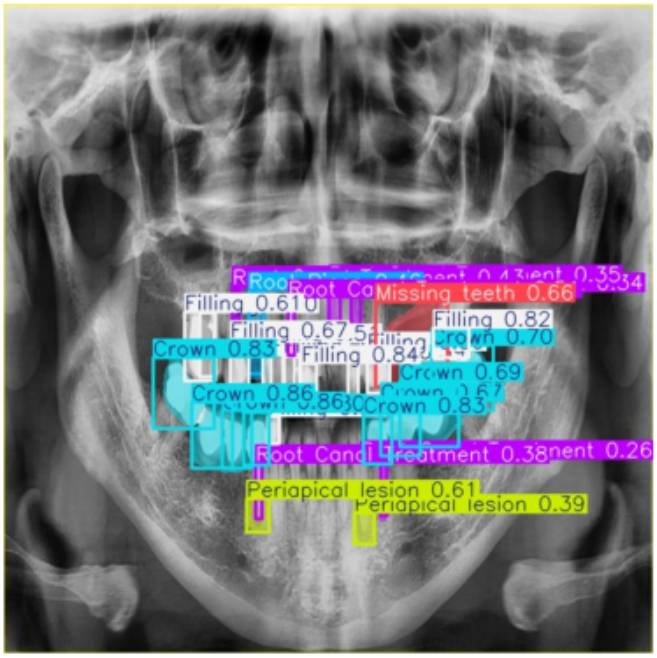
```
image 1/1 /content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/test/images/000dc27f
Speed: 3.3ms preprocess, 110.0ms inference, 79.8ms postprocess per image at shape (1,
Results saved to /content/runs/segment/predict
```

```
!ls runs/segment/predict
```

```
000dc27f-NAJIB_MARDANLOO_MASUME_2020-07-12185357_jpg.rf.adab6a526a5ec0225d2ce97c33095e
```

```
from PIL import Image
import matplotlib.pyplot as plt

result_img = Image.open("runs/segment/predict/000dc27f-NAJIB_MARDANLOO_MASUME_2020-07
plt.imshow(result_img)
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(639.5), np.float64(639.5), np.float64(-0.5))
```




```
for r in results:
  print(r.boxes)
```

```
ultralytics.engine.results.Boxes object with attributes:

cls: tensor([ 1.,  1.,  2.,  1.,  1.,  2.,  1.,  1.,  1.,  1.,  2.,  6.,  7.,  2.,
conf: tensor([0.8642, 0.8606, 0.8378, 0.8341, 0.8255, 0.8246, 0.7993, 0.7022, 0.6917
data: tensor([[1.8578e+02, 3.9281e+02, 2.2558e+02, 4.5761e+02, 8.6418e-01, 1.0000e+0
        [2.1505e+02, 4.0074e+02, 2.4445e+02, 4.5735e+02, 8.6061e-01, 1.0000e+00],
        [2.9372e+02, 3.5410e+02, 3.0898e+02, 3.9346e+02, 8.3784e-01, 2.0000e+00],
        [1.4712e+02, 3.4842e+02, 2.0778e+02, 4.1890e+02, 8.3406e-01, 1.0000e+00],
        [3.5496e+02, 4.0459e+02, 3.8249e+02, 4.5352e+02, 8.2547e-01, 1.0000e+00],
        [4.2400e+02, 3.1980e+02, 4.5806e+02, 3.4881e+02, 8.2463e-01, 2.0000e+00],
        [2.3635e+02, 4.0295e+02, 2.6163e+02, 4.5445e+02, 7.9926e-01, 1.0000e+00],
        [4.2120e+02, 3.3656e+02, 4.8252e+02, 4.0018e+02, 7.0216e-01, 1.0000e+00],
        [3.9017e+02, 3.7169e+02, 4.4835e+02, 4.3614e+02, 6.9166e-01, 1.0000e+00],
        [3.7180e+02, 3.9140e+02, 4.1007e+02, 4.4515e+02, 6.7478e-01, 1.0000e+00],
        [2.2398e+02, 3.3089e+02, 2.4110e+02, 3.8010e+02, 6.7133e-01, 2.0000e+00],
        [3.6636e+02, 2.9344e+02, 4.3744e+02, 3.7746e+02, 6.6361e-01, 6.0000e+00],
        [2.3989e+02, 4.8779e+02, 2.6166e+02, 5.2028e+02, 6.1348e-01, 7.0000e+00],
        [1.7899e+02, 3.0304e+02, 2.0857e+02, 3.7105e+02, 6.0532e-01, 2.0000e+00],
        [3.5855e+02, 3.4187e+02, 3.8027e+02, 3.8387e+02, 5.9348e-01, 2.0000e+00],
        [3.3985e+02, 3.5122e+02, 3.4950e+02, 3.8511e+02, 5.8526e-01, 2.0000e+00],
        [2.6038e+02, 4.0670e+02, 2.7199e+02, 4.3315e+02, 5.5103e-01, 2.0000e+00],
        [2.5576e+02, 3.3318e+02, 2.7513e+02, 3.7956e+02, 5.0830e-01, 2.0000e+00],
        [2.8011e+02, 2.8972e+02, 2.8641e+02, 3.4642e+02, 5.0788e-01, 9.0000e+00],
        [2.7564e+02, 3.4544e+02, 2.8667e+02, 3.8497e+02, 4.9874e-01, 2.0000e+00],
        [2.4115e+02, 2.8204e+02, 2.5606e+02, 3.6422e+02, 4.5694e-01, 1.0000e+01],
        [2.2508e+02, 2.7819e+02, 2.3103e+02, 3.3885e+02, 4.2665e-01, 9.0000e+00],
        [2.7629e+02, 3.4717e+02, 2.9188e+02, 3.8598e+02, 3.9192e-01, 2.0000e+00],
        [3.4505e+02, 5.0122e+02, 3.6510e+02, 5.3266e+02, 3.8970e-01, 7.0000e+00],
        [2.4808e+02, 4.5283e+02, 2.5465e+02, 5.0716e+02, 3.7835e-01, 9.0000e+00],
        [3.2087e+02, 2.7515e+02, 3.3162e+02, 3.4808e+02, 3.5353e-01, 9.0000e+00],
        [3.4228e+02, 2.8394e+02, 3.5159e+02, 3.4962e+02, 3.4073e-01, 9.0000e+00],
        [2.6303e+02, 3.4588e+02, 2.7517e+02, 3.7773e+02, 3.2455e-01, 2.0000e+00],
        [1.9323e+02, 3.0492e+02, 2.0877e+02, 3.6791e+02, 3.0022e-01, 2.0000e+00],
        [2.2598e+02, 2.7633e+02, 2.3294e+02, 3.3930e+02, 2.8645e-01, 9.0000e+00],
        [3.7133e+02, 4.4874e+02, 3.7789e+02, 5.0487e+02, 2.5679e-01, 9.0000e+00]], d
```

```
id: None
is_track: False
orig_shape: (640, 640)
shape: torch.Size([31, 6])
xywh: tensor([[205.6819, 425.2131,  39.8046,  64.7985],
        [229.7506, 429.0425,  29.3997,  56.6136],
        [301.3516, 373.7783,  15.2654,  39.3578],
        [177.4518, 383.6636,  60.6569,  70.4824],
        [368.7249, 429.0540,  27.5314,  48.9373],
        [441.0289, 334.3065,  34.0585,  29.0134],
        [248.9915, 428.7006,  25.2768,  51.5032],
        [451.8588, 368.3694,  61.3198,  63.6285],
        [419.2586, 403.9136,  58.1739,  64.4465],
        [390.9379, 418.2753,  38.2721,  53.7494],
        [232.5384, 355.4949,  17.1208,  49.2088],
        [401.9009, 335.4500,  71.0721,  84.0271],
        [250.7745, 504.0385,  21.7686,  32.4889],
        [193.7799, 337.0448,  29.5798,  68.0017],
        [369.4069, 362.8718,  21.7191,  42.0031],
        [344.6767, 368.1666,   9.6495,  33.8892],
        [266.1837, 419.9215,  11.6115,  26.4525],
        [265.4438, 356.3669,  19.3693,  46.3821],
```

```
for r in results:
  boxes = r.boxes

  for i in range(len(boxes)):
    cls_id = int(boxes.cls[i])
    conf = float(boxes.conf[i])
    coords = boxes.xyxy[i].tolist()

    #Filter low confidence
    if conf > 0.5:
      print(f"Class: {cls_id}, Confidence: {conf:.2f}, Box: {coords}")
```

```
Class: 1, Confidence: 0.86, Box: [185.77963256835938, 392.81390380859375, 225.5841979
Class: 1, Confidence: 0.86, Box: [215.05072021484375, 400.7357177734375, 244.45043945
Class: 2, Confidence: 0.84, Box: [293.71893310546875, 354.099365234375, 308.984313964
Class: 1, Confidence: 0.83, Box: [147.12335205078125, 348.42236328125, 207.7802124023
Class: 1, Confidence: 0.83, Box: [354.9591979980469, 404.58538818359375, 382.49063110
Class: 2, Confidence: 0.82, Box: [423.9996032714844, 319.7998046875, 458.05813598632
Class: 1, Confidence: 0.80, Box: [236.35308837890625, 402.948974609375, 261.62988281
Class: 1, Confidence: 0.70, Box: [421.1988525390625, 336.55511474609375, 482.51867675
Class: 1, Confidence: 0.69, Box: [390.1716613769531, 371.6903076171875, 448.34555053
Class: 1, Confidence: 0.67, Box: [371.8018798828125, 391.400634765625, 410.07397460
Class: 2, Confidence: 0.67, Box: [223.97799682617188, 330.8905029296875, 241.09884643
Class: 6, Confidence: 0.66, Box: [366.3648681640625, 293.43646240234375, 437.43695068
Class: 7, Confidence: 0.61, Box: [239.89019775390625, 487.79400634765625, 261.65875244
Class: 2, Confidence: 0.61, Box: [178.989990234375, 303.0439147949219, 208.5698242187
Class: 2, Confidence: 0.59, Box: [358.54730224609375, 341.8702087402344, 380.26641845
Class: 2, Confidence: 0.59, Box: [339.8519287109375, 351.22198486328125, 349.50146484
Class: 2, Confidence: 0.55, Box: [260.3780212402344, 406.69525146484375, 271.98947143
Class: 2, Confidence: 0.51, Box: [255.75914001464844, 333.17584228515625, 275.12841790
Class: 9, Confidence: 0.51, Box: [280.10723876953125, 289.7154541015625, 286.40570068
```

```
class_names = model.names

for r in results:
    boxes = r.boxes
```

```
    for i in range(len(boxes)):
        cls_id = int(boxes.cls[i])
        conf = float(boxes.conf[i])

        if conf > 0.5:
            print(f"{class_names[cls_id]} → {conf:.2f}")
```

```
Crown → 0.86
Crown → 0.86
Filling → 0.84
Crown → 0.83
Crown → 0.83
Filling → 0.82
Crown → 0.80
Crown → 0.70
Crown → 0.69
Crown → 0.67
Filling → 0.67
Missing teeth → 0.66
Periapical lesion → 0.61
Filling → 0.61
Filling → 0.59
Filling → 0.59
Filling → 0.55
Filling → 0.51
Root Canal Treatment → 0.51
```

```
!cat /content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/data.yaml
```

```
# path: ../datasets/roboflow

train: /home/loki/DentalObjectDetection/data/YOLO/train/images
val: /home/loki/DentalObjectDetection/data/YOLO/valid/images
test: /home/loki/DentalObjectDetection/data/YOLO/valid/images

names:
  0: Caries
  1: Crown
  2: Filling
  3: Implant
  4: Malaligned
  5: Mandibular Canal
  6: Missing teeth
  7: Periapical lesion
  8: Retained root
  9: Root Canal Treatment
  10: Root Piece
  11: impacted tooth
  12: maxillary sinus
  13: Bone Loss
  14: Fracture teeth
  15: Permanent Teeth
  16: Supra Eruption
  17: TAD
  18: abutment
  19: attrition
  20: bone defect
  21: gingival former
  22: metal band
  23: orthodontic brackets
  24: permanent retainer
  25: post - core
  26: plating
```

```
  27: wire
  28: Cyst
  29: Root resorption
  30: Primary teeth
```

```
import yaml

data_yaml_path = "/content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/data.yaml"

with open(data_yaml_path, 'r') as f:
  data = yaml.safe_load(f)

BASE_PATH = "/content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO"
#Update paths
data['train'] = BASE_PATH + "/train/images"
data['val'] = BASE_PATH + "/valid/images"
data['test'] = BASE_PATH + "/test/images"

#save new file
fixed_yaml_path = BASE_PATH + "/data_fixed.yaml"

with open(fixed_yaml_path, 'w') as f:
  yaml.dump(data, f)

print("Fixed YAML saved at:", fixed_yaml_path)
```

```
Fixed YAML saved at: /content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/data_fixe
```

step 2 : START FIRST FINE-TUNING

```
from ultralytics import YOLO

model = YOLO("/content/drive/MyDrive/Dental_Colab_Notebooks/best.pt")

# Train (fine-tune)
model.train(
    data = fixed_yaml_path,
    epochs = 5,
    imgsz = 640,
    batch = 8,
    project = "/content/drive/MyDrive/Dental_Colab_Notebooks/runs",
    name="dental_train",
    save=True
)

print("training completed")
```

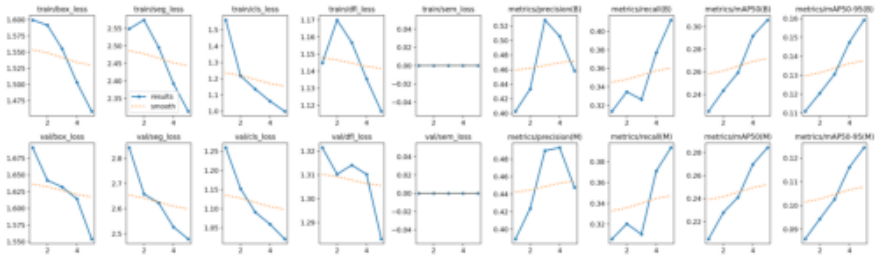
```
Ultralytics 8.4.37 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment
Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/A

                   from  n    params  module                                       a
  0                  -1  1      2320  ultralytics.nn.modules.conv.Conv             [
  1                  -1  1    115520  ultralytics.nn.modules.conv.Conv             [
  2                  -1  3    436800  ultralytics.nn.modules.block.C2f             [
  3                  -1  1    461440  ultralytics.nn.modules.conv.Conv             [
```

```
  4                  -1  6   3281920  ultralytics.nn.modules.block.C2f             [
  5                  -1  1   1844480  ultralytics.nn.modules.conv.Conv             [
  6                  -1  6  13117440  ultralytics.nn.modules.block.C2f             [
  7                  -1  1   3687680  ultralytics.nn.modules.conv.Conv             [
  8                  -1  3   6969600  ultralytics.nn.modules.block.C2f             [
  9                  -1  1   1025920  ultralytics.nn.modules.block.SPPF            [
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample         [
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [
 12                  -1  3   7379200  ultralytics.nn.modules.block.C2f             [
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample         [
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [
 15                  -1  3   1948800  ultralytics.nn.modules.block.C2f             [
 16                  -1  1    922240  ultralytics.nn.modules.conv.Conv             [
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat           [
 18                  -1  3   7174400  ultralytics.nn.modules.block.C2f             [
 19                  -1  1   3687680  ultralytics.nn.modules.conv.Conv             [
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat           [
 21                  -1  3   7379200  ultralytics.nn.modules.block.C2f             [
 22        [15, 18, 21]  1  12346061  ultralytics.nn.modules.head.Segment          [
YOLOv8x-seg summary: 232 layers, 71,780,701 parameters, 71,780,685 gradients, 328.9

Transferred 621/657 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26n.p
AMP: checks passed ✅
train: Fast image access ✅ (ping: 184.5±173.5 ms, read: 0.1±0.0 MB/s, size: 72.1 KB
train: Scanning /content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/train/labels
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7
val: Fast image access ✅ (ping: 0.7±0.2 ms, read: 0.2±0.1 MB/s, size: 63.9 KB)
val: Scanning /content/drive/MyDrive/Dental_Colab_Notebooks/YOLO/YOLO/valid/labels.c
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and dete
optimizer: AdamW(lr=0.000286, momentum=0.9) with parameter groups 106 weight(decay=0
Plotting labels to /content/drive/MyDrive/Dental_Colab_Notebooks/runs/dental_train7/
Image sizes 640 train, 640 val
Using 2 dataloader workers
Logging results to /content/drive/MyDrive/Dental_Colab_Notebooks/runs/dental_train7
Starting training for 5 epochs...

      Epoch    GPU_mem   box_loss   seg_loss   cls_loss   dfl_loss   sem_loss  Insta
        1/5      7.84G        1.6      2.549      1.561      1.145          0
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
                   all       2871      27141      0.403      0.314      0.225      0

      Epoch    GPU_mem   box_loss   seg_loss   cls_loss   dfl_loss   sem_loss  Insta
        2/5      7.76G      1.592      2.574      1.218       1.17          0
                 Class     Images  Instances      Box(P          R      mAP50  mAP50
                   all       2871      27141      0.433      0.335      0.243      0
```

RESUME TRAINING

```
"""import os
from ultralytics import YOLO

base = "/content/drive/MyDrive/Dental_Colab_Notebooks/runs"

folders = sorted(os.listdir(base))
latest = folders[-1]

checkpoint = f"{base}/{latest}/weights/last.pt"
```

```
if os.path.exists(checkpoint):
    print("🔁 Resuming training from checkpoint...")
    model = YOLO(checkpoint)
    model.train(resume=True)
else:
    print("🚀 Starting fresh training...")
    model = YOLO("/content/drive/MyDrive/Dental_Colab_Notebooks/best.pt")
    model.train(
        data=fixed_yaml_path,
        epochs=5,
        imgsz=640,
        batch=8,
        project="/content/drive/MyDrive/Dental_Colab_Notebooks/runs",
        name="dental_train7",
        save=True
    )"""
```

```
from PIL import Image
import matplotlib.pyplot as plt

img = Image.open("/content/drive/MyDrive/Dental_Colab_Notebooks/runs/dental_train7/re

plt.imshow(img)
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(3999.5), np.float64(1199.5), np.float64(-0.5))
```